MINI-PROJECT REPORT

INDUSTRIAL 2-FINGER GRIPPER
– CAD DESIGN & STRESS SIMULATION

Parametric CAD Design and Static Structural Validation of a Rack-and-Pinion Parallel-Jaw Industrial Gripper

Prepared by	:	Aswin S Pillai
CAD & Simulation Software	:	Autodesk Fusion 360 (version 2705.1.15)
Simulation Study	:	Study 1 – Static Stress (completed 23 September 2026)
Design Payload	:	8 kg

ABSTRACT

Industrial robotic systems require end-effectors capable of securely holding and manipulating objects. This mini-project presents the parametric CAD design and static structural validation of a two-finger parallel-jaw industrial gripper that uses a rack-and-pinion actuation mechanism. A digital servo drives a 20-tooth, module-1 pinion, which engages two opposing racks to produce symmetric, parallel motion of the two gripping jaws, with steel guide rods maintaining alignment. The gripper was sized for an 8 kg payload. Assuming a coefficient of friction of 0.5 and a design safety factor of 1.5, the payload weight of 78.48 N gives a theoretical minimum gripping force of 117.72 N per jaw; a rounded design value of 120 N per jaw was adopted and used directly as the applied load in the simulation. The CAD model was developed in Autodesk Fusion 360, and a static structural (Study 1 – Static Stress) simulation was performed with the robotic-arm mounting interface fixed, gravity active, and 120 N applied to each jaw. The study reports a minimum safety factor of 11.98, a maximum von Mises stress of 17.279 MPa, a maximum total displacement of 0.003 mm and a maximum equivalent strain of 1.568×10^{-4}, all obtained directly from the Fusion 360 Study Report. No physical prototype testing has been performed; the results represent simulation-based validation only.

Keywords: parallel-jaw gripper; rack and pinion; parametric CAD; Autodesk Fusion 360; static stress analysis; safety factor.

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1. INTRODUCTION

Industrial robotic systems require end-effectors capable of securely holding and manipulating objects during pick-and-place, assembly and handling tasks. A parallel-jaw gripper is one of the most common mechanisms used for this purpose, because two opposing jaws can move toward and away from each other while keeping their gripping faces approximately parallel.

This project develops a two-finger parallel-jaw industrial gripper using a rack-and-pinion actuation mechanism. A digital servo provides rotary motion, which a pinion converts into linear motion of two opposing racks. The racks drive the two gripping jaws symmetrically, so that the jaws open and close together while remaining aligned by a pair of guide rods. The gripper was sized for an 8 kg payload.

The work covers the full design-to-validation chain: concept development, parametric CAD modeling in Autodesk Fusion 360, dimensioning, assembly, gripping-force calculation, and a Fusion 360 static structural (Static Stress) simulation used to evaluate stress, displacement, strain and safety factor against the project requirement of a minimum safety factor of 1.5.

2. OBJECTIVES

The objectives of the project are:

1. To design a two-finger parallel-jaw industrial gripper suitable for robotic applications.
2. To develop the gripper using parametric CAD modeling in Autodesk Fusion 360.
3. To implement a rack-and-pinion mechanism that converts servo rotation into symmetric linear jaw motion.
4. To design the gripper for an 8 kg payload.
5. To calculate the required gripping force, accounting for friction and a design safety factor.
6. To assemble the components correctly and verify jaw symmetry and parallel motion.
7. To perform a static structural (Static Stress) simulation of the gripper in Autodesk Fusion 360.
8. To evaluate the resulting stress, displacement, strain and safety factor.
9. To verify whether the design satisfies the required minimum safety factor of 1.5.

3. DESIGN REQUIREMENTS

The mini-project is required to satisfy the following design requirements:

- A two-finger parallel-jaw gripper with fingers, a base structure and an actuation interface, suitable for robotic applications.
- Parametric CAD modeling of all components, correctly assembled.
- A design payload of 8 kg.
- Demonstrated jaw symmetry and parallel jaw movement.
- A calculated gripping force that accounts for friction and includes a design safety factor.
- A static structural simulation that reports stress, strain, displacement and safety factor.
- A minimum structural safety factor of 1.5.

These requirements are addressed in the corresponding sections of this report: the CAD design in Sections 6–12, the mechanism and actuation in Sections 11–12, materials in Section 13, the gripping-force calculation in Section 14, and the static structural simulation and its results in Sections 15–20.

4. CONCEPT DEVELOPMENT

4.1 Selected mechanism

The selected mechanism is a two-finger parallel-jaw gripper driven by a rack-and-pinion arrangement. A single rotary actuator drives a central pinion that meshes with two racks placed on opposite sides of the gripper body. Because the two racks are designed to move equal distance in opposite directions, one actuator is sufficient to drive both jaws symmetrically.

4.2 Operating principle

The operating sequence of the gripper is as follows:

1. The servo provides rotary motion at its output shaft.
2. The servo output is connected to the pinion shaft through a coupler.
3. The coupler rotates the pinion.
4. The pinion engages with two opposing racks.
5. Rotation of the pinion produces equal and opposite linear movement of the two racks.
6. Each rack forms part of one gripping jaw.
7. The two jaws therefore move toward each other when closing and away from each other when opening.
8. A pair of guide rods constrains the jaws to translate along the gripping axis and keeps them aligned.
9. Rubber jaw pads mounted on the jaws contact the object and generate the frictional gripping force.

4.3 Parallel motion and symmetry

Because both jaw carriers are constrained to translate along the same guide-rod axis while being driven in opposite directions by the shared pinion, the two jaw faces remain approximately parallel throughout the stroke and the gripped object tends to remain centred between the jaws.

5. INITIAL CONCEPT / HAND SKETCH

As is normal design practice, the rack-and-pinion parallel-gripper concept described in Section 4 was first developed as a hand-drawn concept sketch before detailed CAD modeling began.

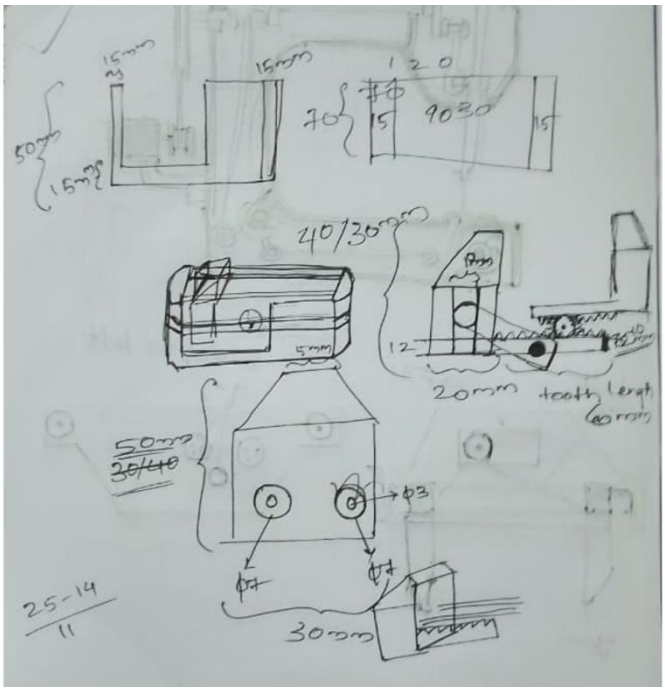


Figure 0. Initial hand sketch.

During subsequent CAD development in Autodesk Fusion 360, the concept was refined to obtain a practical rack-and-pinion mechanism, appropriate mounting interfaces, correct component integration and a structure suitable for parametric modelling and simulation. The resulting CAD design is presented in Section 6 onward.

6. FINAL CAD DESIGN

6.1 Modelling environment and components

The final gripper assembly was modelled in Autodesk Fusion 360. The major components of the assembly, as listed in the Fusion 360 model browser, are summarised in the table below.

Table 1. Major components of the gripper assembly (Fusion 360 model browser)

Component (browser name)	Function
gripper_base	Main structural body and mounting interface for the mechanism
jaw : 1 / jaw 2 : 1	Combined rack-and-jaw bodies; each carries one gripping jaw and translates on the guide rods
pinion	20-tooth, module-1 spur gear that drives both racks simultaneously in opposite directions
pinion_holder	Supports and locates the pinion / pinion shaft on the base
rod : 1 / rod 2 : 1	Guide rods that constrain jaw motion to the gripping axis
coupler	Connects the servo output to the pinion shaft
DS Servo DS5160 SSG	Digital metal-gear servo that provides the rotary actuation input
jaw_pad : 1 / jaw_pad (1) : 1	Rubber contact pads mounted on the jaws that generate the gripping friction force
arm_connection	Mounting / connection interface to the robotic arm

screws	Fasteners used to assemble the components

6.2 CAD views

The isometric view of the complete assembly is shown in the figure below.



Figure 1. Isometric view of the final gripper assembly.

The front view, shown below, illustrates the two parallel jaws and their symmetric arrangement about the central axis.



Figure 2. Front view of the gripper, showing the two parallel jaws.

The top view, shown below, illustrates the overall footprint of the gripper and the internal rack-and-pinion cavity.

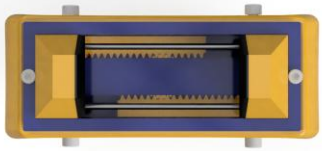


Figure 3. Top view of the gripper assembly.

7. DESIGN DIMENSIONS

The final design dimensions of the gripper are given in the table below.

Table 2. Final design dimensions of the gripper

Parameter	Value
Jaw thickness	20 mm per jaw
Maximum opening between jaw faces	72 mm
Inner jaw-supporting length	120 mm
Approximate travel of each jaw from centre position	36 mm
Overall body length	143 mm
Overall body width	140.2 mm
Base thickness	6 mm
Side wall thickness	6 mm
Side wall height	29 mm
Guide rod diameter	3 mm
Guide rod length	130 mm
Jaw approximate dimensions	81 × 39.2 × 50 mm
Gear / pinion thickness	10 mm
Actuation shaft diameter	6 mm

For symmetric movement, each jaw travels half of the maximum opening from the centre position:

$$s = \frac{g_{\max}}{2} = \frac{72}{2} = 36 \text{ mm} \tag{1}$$

The jaw opening was also verified directly on the CAD assembly using Fusion 360's measurement tool, shown in the figure below, which reports a maximum opening of 72.418 mm — consistent with the 72 mm design value.

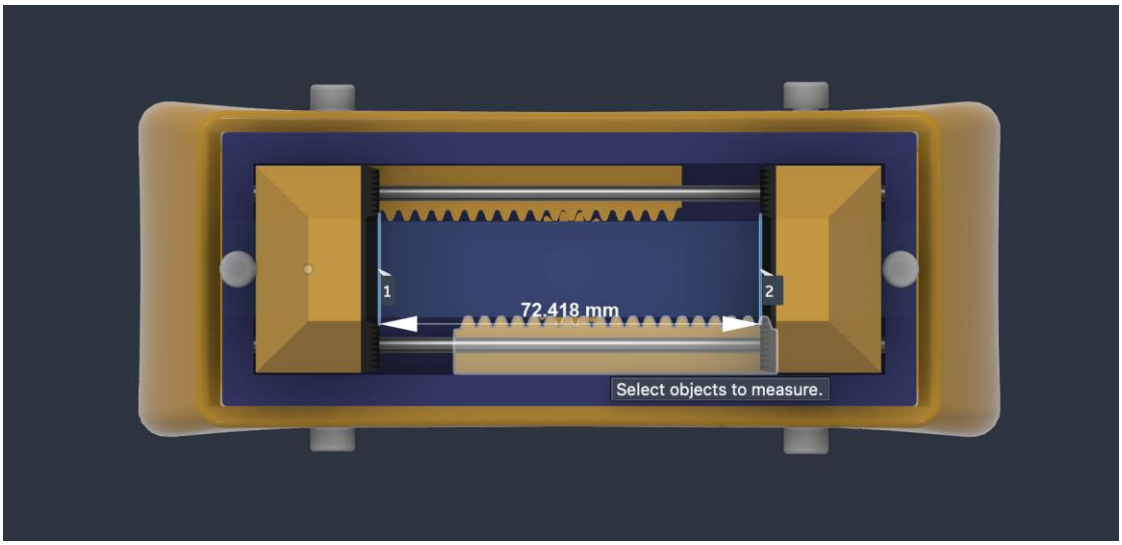


Figure 4. Measured maximum jaw opening on the CAD assembly (72.418 mm), consistent with the 72 mm design value.

> **Note on the inner jaw-supporting length:** the sum of the two 20 mm jaw thicknesses and the 72 mm maximum opening is 20 + 72 + 20 = 112 mm, which is 8 mm less than the 120 mm inner jaw-supporting length used in the design. This additional 8 mm reflects internal clearance in the mechanism (such as rack end-of-travel allowance and the pinion/guide-rod housing) beyond the minimum footprint of the jaws and the opening, rather than a simple additive relationship between the two dimensions.

8. PARAMETRIC CAD DESIGN

In parametric modelling, each feature is driven by sketches and named dimensions rather than fixed geometry, so that a dimension change updates all dependent features consistently. The gripper was modelled in Autodesk Fusion 360 using the following procedure.

8.1 Base and side walls

The gripper_base body was created from a parametric sketch and extruded to form the structural base and side walls (6 mm base thickness, 6 mm side-wall thickness, 29 mm side-wall height), which provide the mounting interface for the mechanism and the arm connection.

8.2 Gripping jaws / racks

Each gripping jaw was modelled as a combined rack-and-jaw body (components rack : 1 and rack 2 : 1), so that the rack teeth and the jaw structure form a single part per side. The two bodies were modelled symmetrically about the gripper centreline.

8.3 Guide rods

Two 3 mm diameter, 130 mm long guide rods (rod : 1 and rod 2 : 1) were added to constrain the jaw bodies to translate along the gripping axis and to maintain their alignment during opening and closing.

8.4 Pinion

The pinion was generated using Fusion 360's spur gear modelling tool, with the parameters summarised below.

Table 3. Pinion parameters

Parameter	Value
Number of teeth, Z	20
Module, m	1 mm
Pressure angle	20°
Gear / pinion thickness	10 mm
Pitch diameter, $D = mZ$	20 mm
Pitch radius, r	10 mm
Root diameter (measured on the fully constrained sketch)	17.686 mm
Bore / shaft interface diameter	6 mm

The pitch diameter follows from the module and the number of teeth:

$$D = mZ = 1 \times 20 = 20 \text{ mm}, \qquad r = \frac{D}{2} = 10 \text{ mm} \tag{2}$$

8.5 Coupler and actuation shaft

The coupler connects the servo output to the pinion through a 6 mm diameter actuation shaft interface, matching the 6 mm bore visible on the pinion's fully constrained sketch (Section 9).

8.6 Fillets and chamfers

Fillets and chamfers were applied where appropriate to reduce sharp edges, improve manufacturability and reduce local stress concentration at suitable locations. It is not claimed that every fillet or chamfer was individually optimised through stress analysis.

9. FULLY CONSTRAINED SKETCHES

A fully constrained Fusion 360 sketch has its geometry and position completely defined by dimensions and geometric constraints, which improves parametric stability and prevents unintended changes when the model is later modified. The base, jaw and pinion sketches were fully constrained, as shown below.

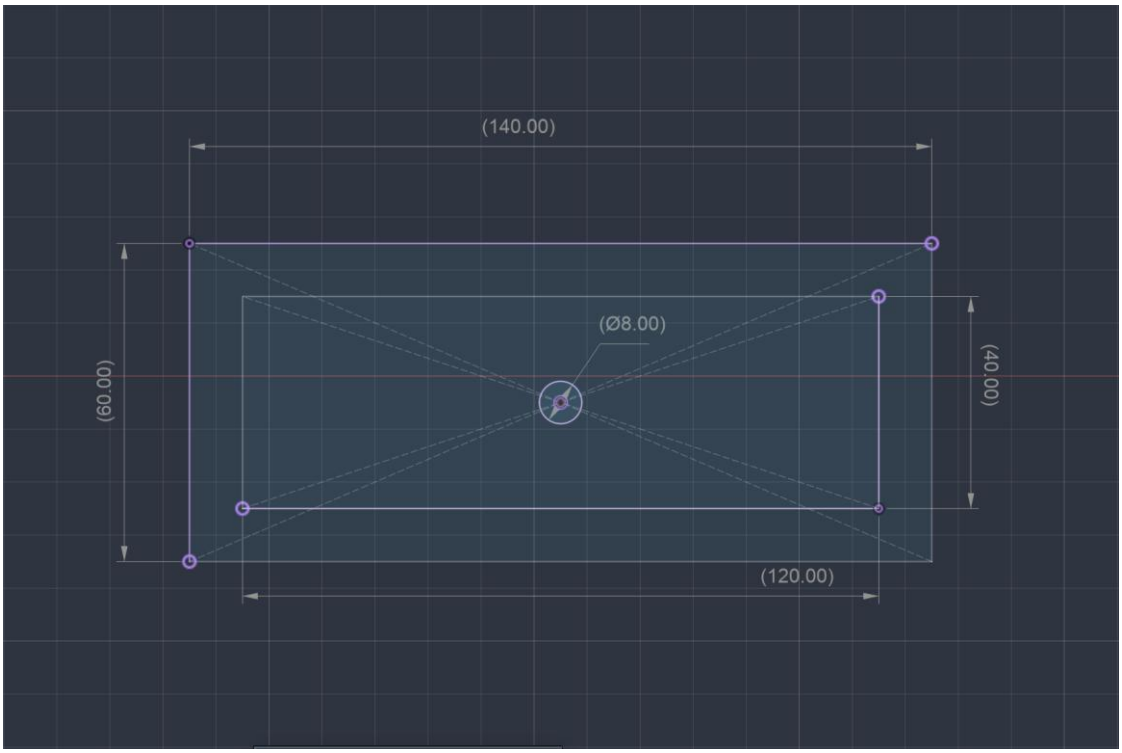


Figure 5. Base sketch, fully constrained, with an outer profile of 140.00 × 60.00 mm, an inner rectangle of 120.00 × 40.00 mm and a central Ø8.00 mm hole. The 120.00 mm inner dimension corresponds to the inner jaw-supporting length in Table 2.

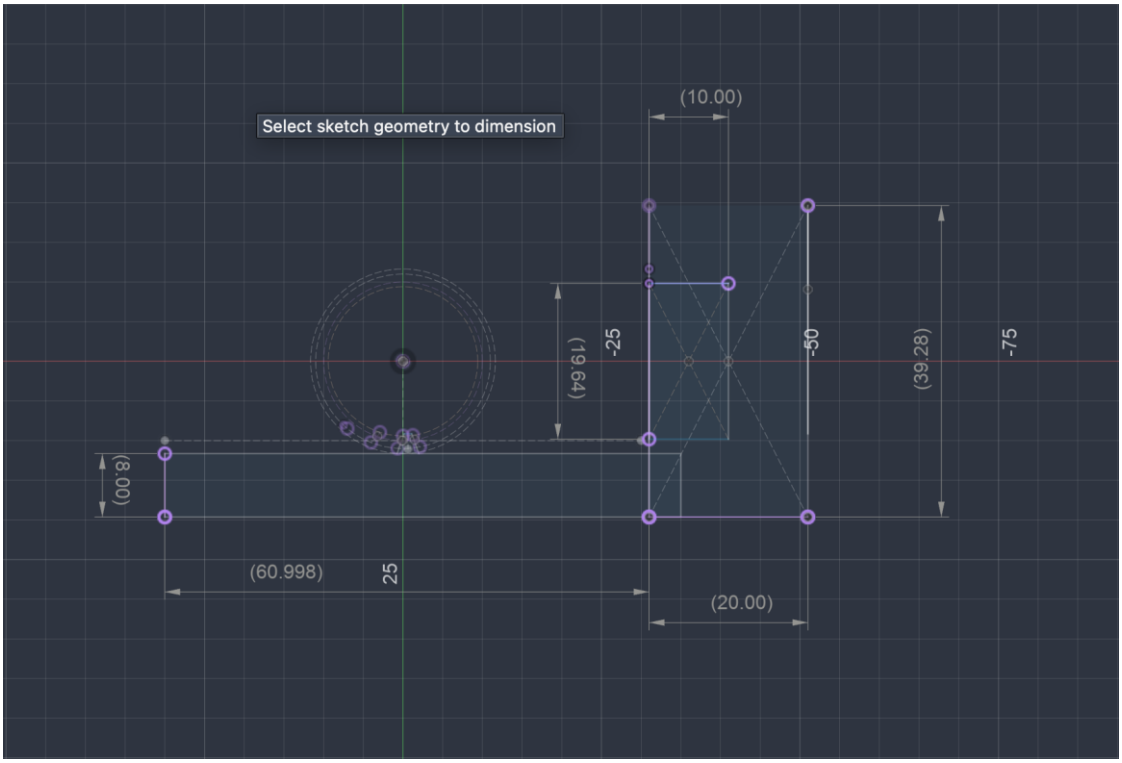


Figure 6. Jaw / rack sketch, fully constrained. The dimensioned jaw height of 39.28 mm is consistent with the 39.2 mm jaw dimension used in the design.

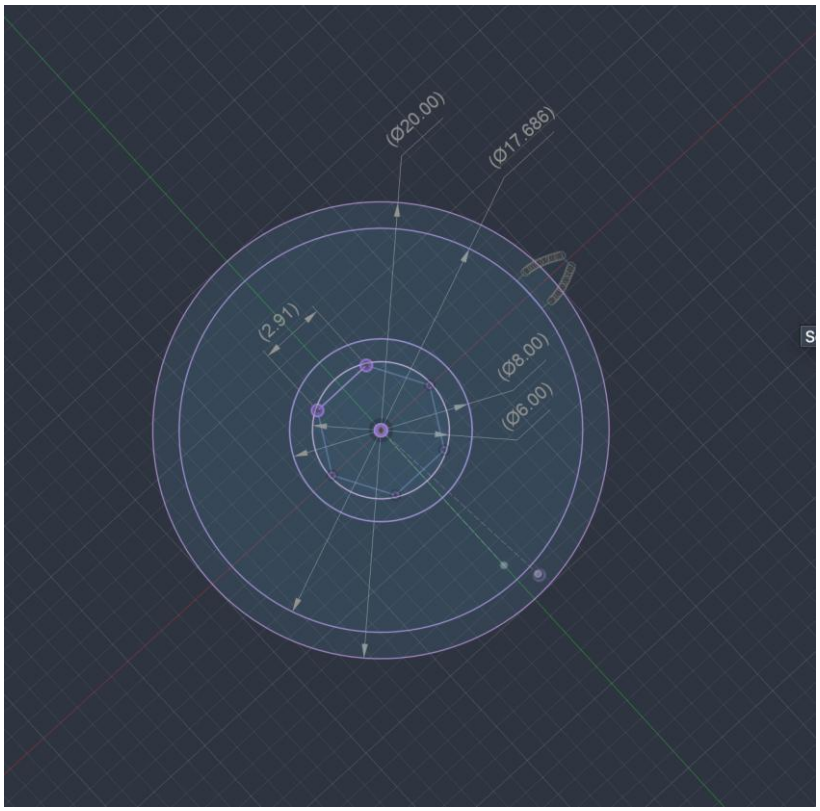


Figure 7. Pinion sketch, fully constrained, showing the Ø20.00 mm pitch diameter, Ø17.686 mm root diameter and Ø6.00 mm bore for the actuation shaft.

10. ASSEMBLY

The complete assembly, together with the Fusion 360 model browser listing every component, is shown below.



Figure 8. Fusion 360 model browser and isometric view of the complete gripper assembly.

The two jaw / rack bodies are arranged symmetrically about the central pinion, and the guide rods constrain their motion to the gripping axis so that the racks are designed to be displaced equally and oppositely by pinion rotation. As established in Section 7, each jaw travels approximately 36 mm from

the centre position for the 72 mm maximum opening, so the gripped object remains approximately centred between the jaws throughout the stroke.

11. RACK-AND-PINION MECHANISM

The rack-and-pinion mechanism, isolated from the rest of the assembly, is shown below together with the servo that drives it.

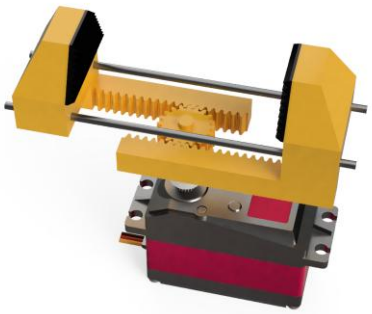


Figure 9. Rack-and-pinion mechanism with the servo actuator; the pinion engages both racks simultaneously.

A second view, with the outer housing removed, more clearly shows the guide rods and the jaw arrangement relative to the pinion.

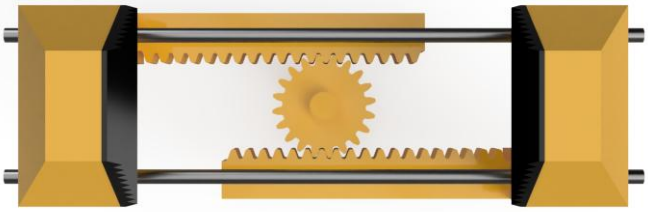


Figure 10. Top view of the rack-and-pinion mechanism with the guide rods and jaws visible; the housing is hidden for clarity.

Pinion rotation is converted to jaw travel through the pitch radius established in Section 8. For the full 36 mm travel of each jaw, the corresponding pinion rotation is:

$$\theta = \frac{s}{r} = \frac{36}{10} = 3.6 \text{ rad} \approx 206^\circ \tag{3}$$

This is a purely geometric relationship based on the pitch radius and does not include any mechanical losses.

12. ACTUATION SYSTEM

The actuation chain, as modelled in the assembly, is: DS5160 SSG digital metal-gear servo → coupler (steel, 6 mm shaft interface) → pinion (steel, 6 mm bore) → pinion_holder (aluminium, provides bearing support) → racks / jaws.

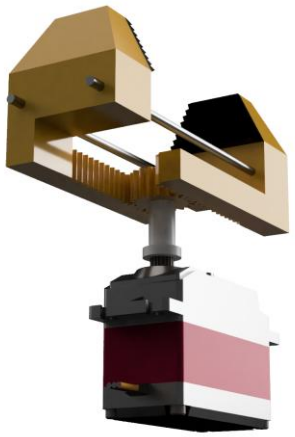


Figure 11. Servo, coupler and pinion / pinion-holder sub-assembly that provides the rotary actuation input.

The pinion_holder supports the pinion shaft on the gripper base. Supporting the pinion shaft in this way reduces the radial load that the rack-and-pinion mesh would otherwise transmit directly to the servo output shaft and coupler.

As calculated in Section 11, the pinion must rotate through approximately 3.6 rad (≈ 206°) to move each jaw through its full 36 mm travel. This rotation range should be verified against the usable travel of the DS5160 SSG servo before the mechanism is built, since it is not confirmed by the files supplied for this report.

13. MATERIAL SELECTION

Material assignments and properties for the static structural simulation were taken directly from the Fusion 360 Study Report (Study Materials), and are reproduced exactly in the table below. Two aluminium entries appear because Fusion's material library lists a generic "Aluminium" (assigned to the fastener bodies) separately from the more specific "Aluminium 6061-T6, O, Cold Formed" (assigned to the main structural bodies).

Table 4. Material assignments and properties used in the static structural simulation (source: Fusion 360 Study Report)

Material	Assigned components	Density	Young's modulus	Poisson's ratio	Yield strength	UTS
Aluminium (generic)	Fastener screws (hex-socket cylindrical-head M4×20, M5×08)	2700 kg/m³	68 900 MPa	0.33	275 MPa	310 MPa
Aluminium 6061-T6, O, Cold Formed	arm_connection, gripper_base, jaw1, jaw2, pinion_holder	2700 kg/m³	69 000 MPa	0.33	369 MPa	389 MPa

Steel (generic)	coupler, pinion, rod, rod 2	7850 kg/m³	210 000 MPa	0.30	207 MPa	345 MPa
Rubber, Butyl	jaw_pad, jaw_pad (1)	120 kg/m³	2.00 MPa	0.49	15 MPa	15 MPa

The material-assignment screenshot from Fusion 360, listing each component and its assigned study material, is shown below.

Component Path	Study Material
Spur Gear 20 teeth:1	Steel
rack:1	Aluminum 6061 T6 0 Cold Formed
rack2:1	Aluminum 6061 T6 0 Cold Formed
gear holder:1	Aluminum 6061 T6 0 Cold Formed
gripper_base:1	Aluminum 6061 T6 0 Cold Formed
rod:1	Steel
rod 2:1	Steel
coupler:1	Steel
jaw_pad:1	Rubber, Butyl
jaw_pad (1):1	Rubber, Butyl
arm_connection:1	Aluminum 6061 T6 0 Cold Formed

Figure 12. Material assignment table from the Fusion 360 Study Materials panel.

In summary, Aluminium 6061-T6 is used for the main structural body, the two gripping jaws and the pinion support and arm-connection interface; Steel is used for the pinion, the two guide rods and the coupler; and Rubber, Butyl is used for the jaw pads that contact the payload.

14. GRIPPING FORCE CALCULATION

14.1 Payload weight

The design payload is $m = 8$ kg and the acceleration due to gravity is taken as $g = 9.81$ m/s². The payload weight is:

$$W = m\,g = 8 \times 9.81 = 78.48 \text{ N} \tag{4}$$

14.2 Assumptions

- Payload = 8 kg
- Static gravitational loading
- Coefficient of friction, μ = 0.5 (assumed)

- Design safety factor, SF = 1.5
- Two-jaw friction grip
- No additional acceleration or impact load included
- Object geometry and centre-of-mass effects are not separately modelled

14.3 Required gripping force

In a two-jaw friction grip, each jaw contacts the object with a normal force F and can develop a friction force of up to μF. The combined friction force from the two jaws must support the payload weight multiplied by the safety factor:

$$2\mu F \geq \mathrm{SF} \times W \tag{5}$$

Rearranging for the required force per jaw and substituting the values above:

$$F \geq \frac{\mathrm{SF} \times W}{2\mu} = \frac{1.5 \times 78.48}{2 \times 0.5} = 117.72\ \mathrm{N} \tag{6}$$

The theoretical minimum gripping force is therefore **117.72 N per jaw**. A rounded design value of **120 N per jaw** was adopted, giving a total opposing force from the two jaws of:

$$F_{\mathrm{total}} = 2F_{\mathrm{design}} = 2 \times 120 = 240\ \mathrm{N} \tag{7}$$

> The 120 N value is the design gripping force applied per jaw under the stated assumptions. This same value (120 N, applied to each jaw pad contact face in opposing directions) was used directly as the applied load in Load Case 1 of the Fusion 360 static stress simulation described in Section 15, so the hand calculation and the simulated load case are consistent.

14.4 Simplified pinion torque estimate

For the 20-tooth, module-1 pinion, the pitch radius is r = 10 mm = 0.010 m. Using the simplified total rack-force estimate of 240 N from above, the pinion torque is:

$$T = F_{\mathrm{total}}\, r = 240 \times 0.010 \approx 2.4\ \mathrm{N \cdot m} \tag{8}$$

This is a simplified static estimate that does not include gear efficiency, bearing friction, rack friction, servo transmission losses, dynamic acceleration, impact loading or manufacturing tolerances, and must not be interpreted as the exact operating torque required from the servo.

15. STATIC STRESS SIMULATION SETUP

A static structural analysis (Study 1 – Static Stress) was carried out in Autodesk Fusion 360 on the Simulation Model derived from the assembly, comprising the gripper_base, the two jaw bodies, the two guide rods, the pinion, the pinion_holder, the coupler, the two jaw pads, the arm_connection interface and the fastener screws.

Table 5. Summary of the static stress study set-up

Item	Setting
Software	Autodesk Fusion 360 (version 2705.1.15)
Study type / name	Static Stress – Study 1 – Static Stress
Material assignment	As given in Table 4 (Section 13), taken from the Fusion 360 Study Report

Boundary condition	Fixed constraint (Fixed3: Ux, Uy, Uz all restrained) at the robotic-arm mounting interface
Gravity	Fusion 360 default gravity active, 9.807 m/s² in the –Z direction
Applied load	Two 120 N forces (Force1, Force2), one on each jaw pad contact face, acting in opposing directions

Detailed contact definitions are provided in the attached Fusion 360 Study Report (see Section 25, Appendix). The finite element mesh used for the analysis is shown below; full mesh statistics are given in Table 7 (Section 25).

The fixed constraint applied at the arm-mounting interface is shown below.

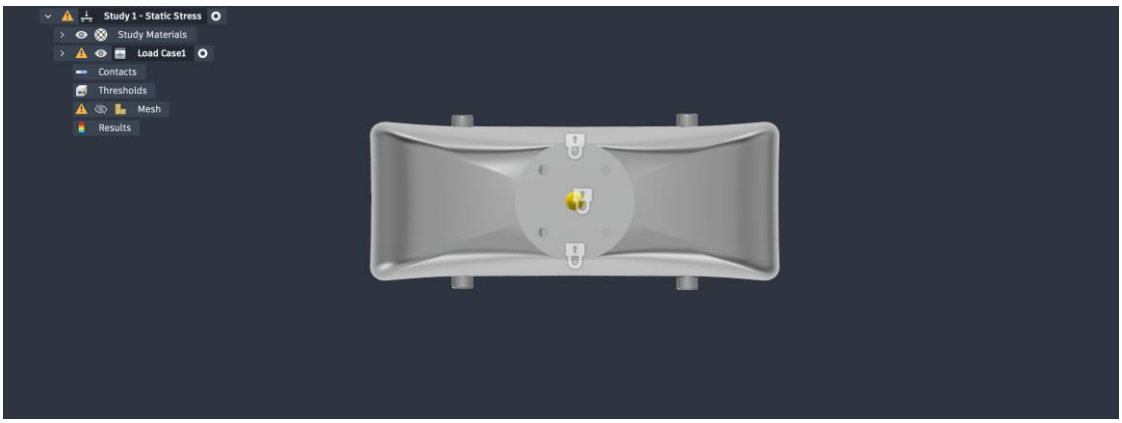


Figure 13. Fixed constraint (Fixed3) applied to the robotic-arm mounting interface at the base of the gripper.

The applied gripping load of 120 N per jaw, together with the fixed constraint, is shown below.

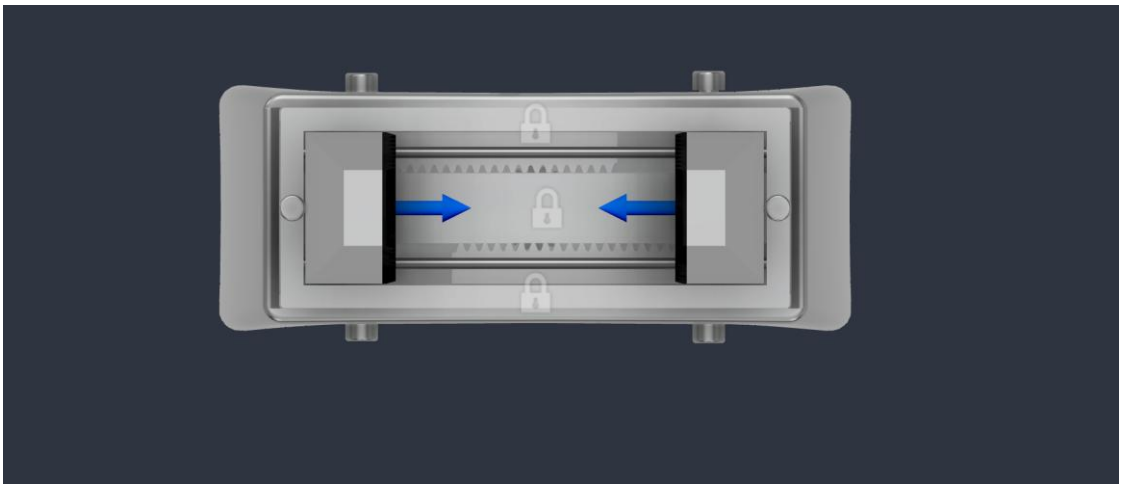


Figure 14. Applied gripping load of 120 N on each jaw pad contact face (Force1 and Force2), together with the fixed constraint at the base.

The finite element mesh used for the analysis (526,544 nodes; 332,937 parabolic solid elements) is shown below.

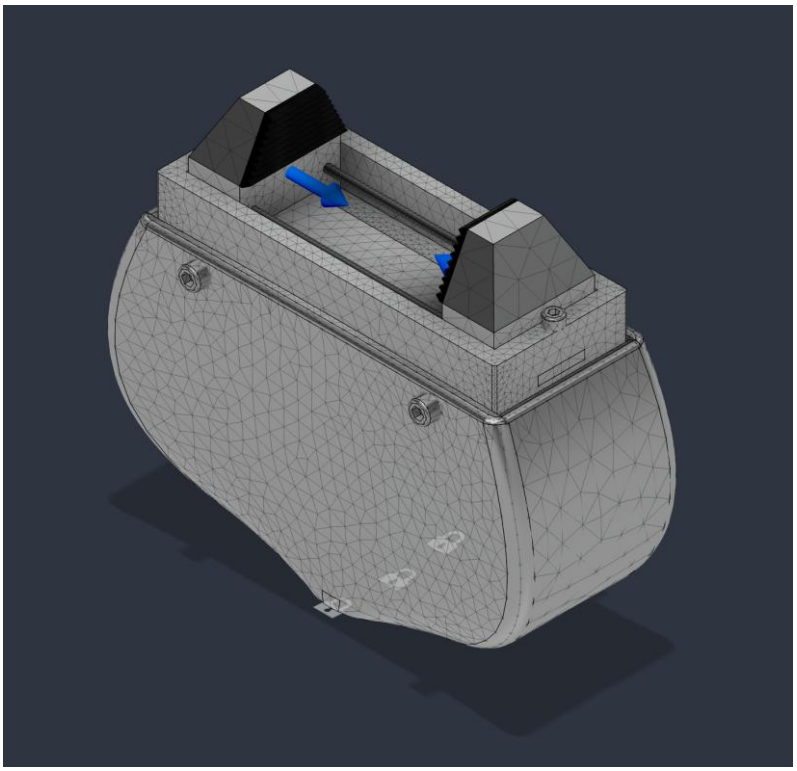


Figure 15. Finite element mesh used for the static stress analysis (526,544 nodes; 332,937 elements).

16. SIMULATION RESULTS

The key results of the static structural simulation, taken directly from the Fusion 360 Study Report, are summarised below and discussed in detail in Sections 17–20.

Table 6. Summary of static structural simulation results

Result	Value	Remarks
Minimum safety factor	11.98	Yield-strength criterion; required ≥ 1.5
Maximum von Mises stress	17.279 MPa	Minimum 2.608×10^{-7} MPa
Maximum total displacement	0.003 mm	Minimum 0 mm (at the fixed support)
Maximum equivalent strain	1.568×10^{-4}	Minimum 8.33×10^{-11}

17. SAFETY FACTOR

Fusion 360 computes the safety factor from the relationship between material strength and calculated stress. For a yield-strength criterion, it is the yield strength divided by the equivalent von Mises stress:

$$\mathrm{SF} = \frac{\sigma_y}{\sigma_{\mathrm{vm}}} \tag{9}$$

where σ_y is the yield strength of the material and σ_{vm} is the equivalent von Mises stress at that location; the lowest value over the whole model is reported as the minimum safety factor.

The completed simulation produced a **minimum safety factor of 11.98**, shown below. Fusion 360's own result summary labels this outcome "Very strong", using a default colour-scale upper target of 4.00 (values above 4.00 are shown as "Above Target").

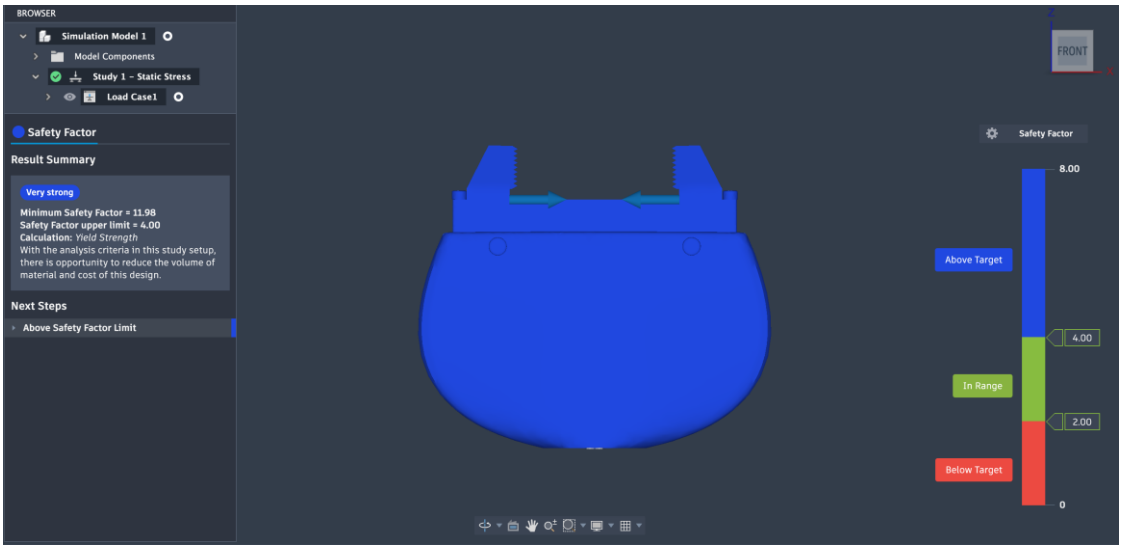


Figure 16. Fusion 360 Safety Factor result: minimum safety factor = 11.98 ("Very strong").

Compared with the project requirement of a minimum safety factor of 1.5:

$$\mathrm{SF}_{\min} = 11.98 > \mathrm{SF}_{\mathrm{req}} = 1.5 \qquad \left(\frac{11.98}{1.5} \approx 7.99\right) \tag{10}$$

> The static structural simulation resulted in a minimum safety factor of 11.98, which is greater than the minimum required safety factor of 1.5. Based on the current model, material assignment, boundary conditions and applied loading, the simulated structure satisfies the specified safety-factor criterion. Fusion 360's own guided results note that, with this safety-factor margin, there is an opportunity to reduce the volume and cost of the design; this is discussed further in Section 21.

18. VON MISES STRESS

Von Mises stress is used to evaluate the yielding tendency of ductile materials, combining the three principal stresses σ_1, σ_2 and σ_3 into a single equivalent stress:

$$\sigma_{\mathrm{vm}} = \sqrt{\frac{1}{2}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]} \tag{11}$$

The von Mises stress distribution obtained from the simulation is shown below.

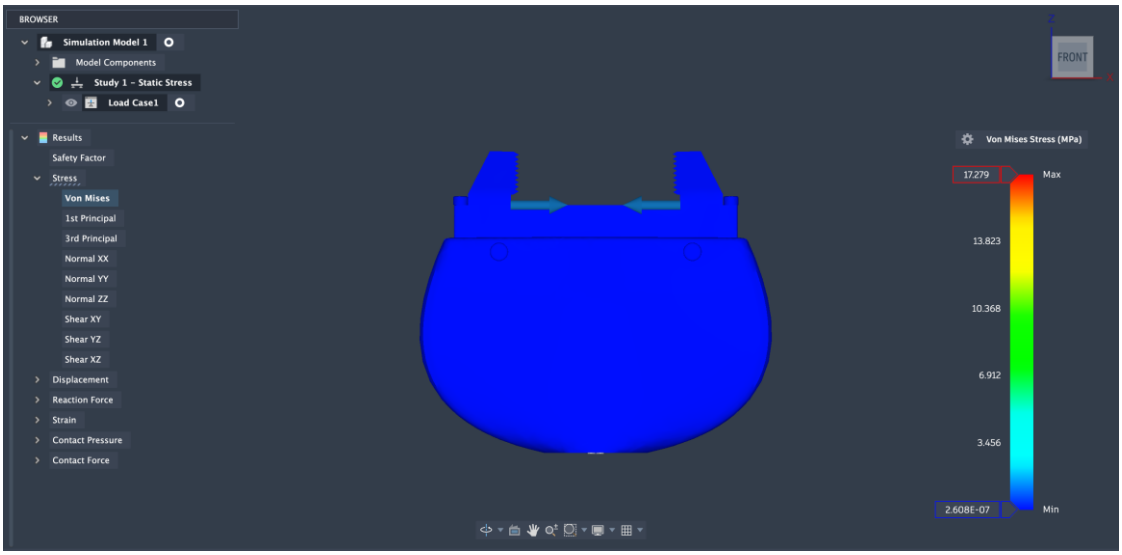


Figure 17. Von Mises stress distribution: maximum = 17.279 MPa, minimum = 2.608×10^{-7} MPa.

The maximum von Mises stress of 17.279 MPa is well below the yield strength of both structural materials used in the model (369 MPa for Aluminium 6061-T6 and 207 MPa for the generic Steel used for the pinion, guide rods and coupler; see Table 4), which is consistent with the high safety factor reported in Section 17.

19. TOTAL DISPLACEMENT

Total displacement represents the movement of points in the model relative to their original position under the applied loading. The result obtained from the simulation is shown below.

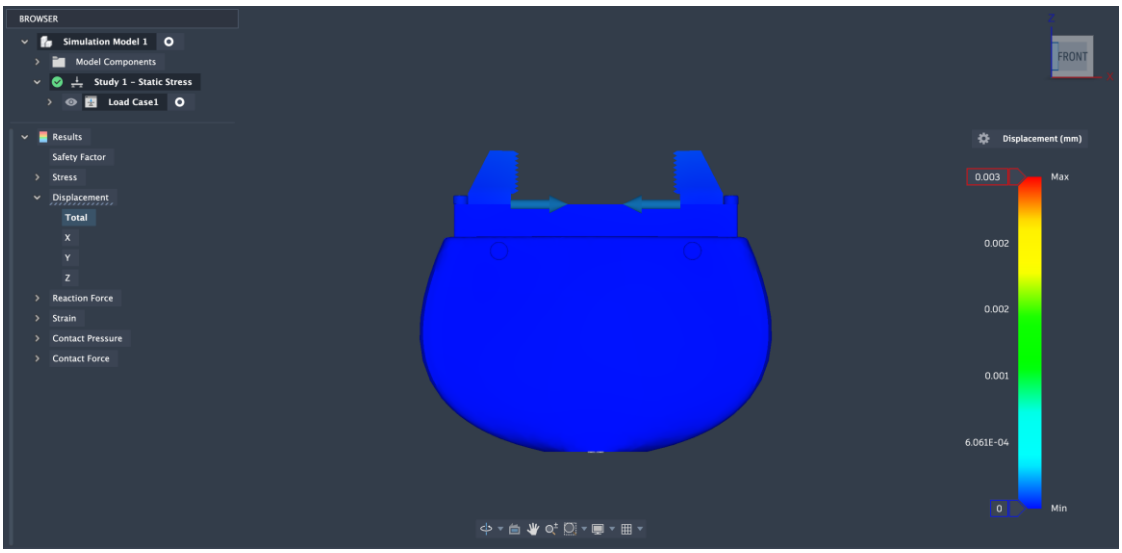


Figure 18. Total displacement distribution: maximum = 0.003 mm, minimum = 0 mm.

The maximum displacement of 0.003 mm occurs at the jaw tips, farthest from the fixed arm-mounting interface, while the displacement is zero at the fixed support itself — consistent with the applied boundary condition. A displacement of this magnitude is negligible relative to the 72 mm jaw opening.

20. EQUIVALENT STRAIN

Strain represents the deformation of the material relative to its original dimensions. For a small change in length ΔL of an element of original length L_0, the normal strain is:

$$\varepsilon = \frac{\Delta L}{L_0} \tag{12}$$

The equivalent strain distribution obtained from the simulation is shown below.

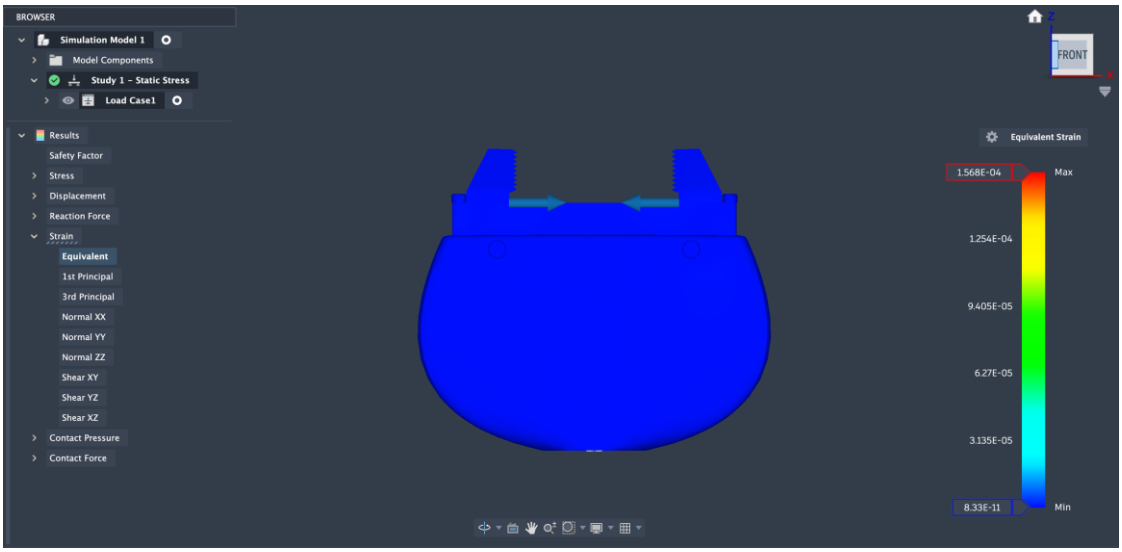


Figure 19. Equivalent strain distribution: maximum = 1.568×10^{-4}, minimum = 8.33×10^{-11}.

The maximum equivalent strain of 1.568×10^{-4} is very small, consistent with the low stress and displacement levels reported above and with the structure remaining well within the elastic range of both structural materials.

21. DISCUSSION

The rack-and-pinion mechanism provides a straightforward way of converting a single servo's rotation into symmetric, opposing jaw motion, while the guide rods maintain jaw alignment throughout the stroke. The 120 N per-jaw design gripping force obtained from the hand calculation in Section 14 was used directly as the applied load in the Fusion 360 simulation, so the structural results in Sections 16–20 are consistent with the stated design assumptions.

The simulation shows a minimum safety factor of 11.98 against a requirement of 1.5, a maximum von Mises stress of 17.279 MPa (well below the yield strength of either structural material used), a maximum displacement of 0.003 mm and a maximum equivalent strain of 1.568×10^{-4}. Fusion 360's own guided results note that, with this large a safety-factor margin, there is an opportunity to reduce material volume and cost in a future revision of the design.

The simulation is, however, a static, idealised analysis and does not represent every real-world operating condition. In particular, the pinion rotation of approximately 206° required for the full jaw travel (Section 12) should be checked against the usable rotation range of the DS5160 SSG servo, since this was not confirmed by the files available for this report. These and other limitations are summarised in Section 22.

22. LIMITATIONS

1. The analysis is static and does not include dynamic robot motion, acceleration or impact loading.
2. The coefficient of friction ($\mu = 0.5$) used in the gripping-force calculation is an assumption, not a measured value for the Rubber, Butyl jaw pads used in the design.
3. Real gripping performance also depends on the payload's surface condition, which is not modelled.
4. Servo efficiency and mechanical transmission losses are not included in the simplified pinion torque calculation.
5. The required pinion rotation of approximately 206° for full jaw travel has not been confirmed against the usable rotation range of the DS5160 SSG servo.
6. Manufacturing tolerances are not evaluated.
7. Fatigue life under repeated gripping cycles is not evaluated.
8. Physical prototype testing has not been performed.
9. The simulation contacts are a mix of automatically generated Bonded, Sliding and Separation contact sets; their full detail and the mesh statistics are documented in the attached Fusion 360 Study Report rather than reproduced in full here.
10. Simulation accuracy depends on the correctness of the material assignments, contacts, constraints and mesh used in the Fusion 360 study.

23. CONCLUSION

This project developed a parametric CAD model of a two-finger, rack-and-pinion parallel-jaw industrial gripper in Autodesk Fusion 360, sized for an 8 kg payload. The rack-and-pinion mechanism converts servo rotation into symmetric, opposing jaw motion, with guide rods maintaining alignment throughout a 72 mm maximum opening (36 mm travel per jaw).

For the 8 kg payload, the calculated payload weight is 78.48 N. With an assumed coefficient of friction of 0.5 and a design safety factor of 1.5, the theoretical minimum gripping force is 117.72 N per jaw; a rounded design value of 120 N per jaw was adopted and used directly as the applied load in the simulation. The corresponding simplified pinion torque estimate is approximately 2.4 N·m.

The Fusion 360 static structural simulation, with the arm-mounting interface fixed and 120 N applied to each jaw pad together with default gravity, produced a minimum safety factor of 11.98, a maximum von Mises stress of 17.279 MPa, a maximum displacement of 0.003 mm and a maximum equivalent strain of 1.568×10^{-4}. Since the minimum safety factor exceeds the required value of 1.5 by a wide margin, the simulated structural model satisfies the project's safety-factor requirement under the stated assumptions.

The design should nevertheless be physically prototyped and tested, and the servo rotation range verified, before use in a real robotic application, since the simulation does not include dynamic, frictional, manufacturing or fatigue effects.

24. REFERENCES

[1] Autodesk Inc., "Static Stress and Structural Analysis," *Autodesk Fusion 360 Help*. Available: https://help.autodesk.com/view/fusion360/ENU/.

[2] Autodesk Inc., "Safety Factor Result," *Autodesk Fusion 360 Help*. Available: https://help.autodesk.com/view/fusion360/ENU/.

[3] Autodesk Inc., "Fully Define and Constrain Sketches," *Autodesk Fusion 360 Help*. Available: https://help.autodesk.com/view/fusion360/ENU/.

[4] Autodesk Fusion 360 Study Report, "Stress study," Study 1 – Static Stress, Simulation Model 1, generated by Autodesk Fusion (version 2705.1.15), 23 September 2026. Supplied as supporting documentation for this report (see Section 25, Appendix).

25. APPENDIX / FUSION 360 STUDY REPORT

This appendix summarises the metadata and setup information recorded in the Fusion 360 Study Report that accompanies this document as a separate supplementary file (Simulation_Studies_Report.html). The complete, interactive study report — including the full list of automatically generated contacts and the mesh statistics — should be read alongside this appendix.

Table 7. Fusion 360 Study Report metadata

Item	Value
Analyzed file	stress study
Fusion 360 version	2705.1.15
Report author	Aswin S Pillai
Study type	Static Stress
Study last modification date	23 September 2026, 12:35:28
Report creation date	23 September 2026, 13:35:44
Mesh type	Solid, parabolic elements
Mesh nodes	526 544
Mesh elements	332 937
Average element size	10% of model size
Contact tolerance	0.10 mm

The contacts between components were automatically generated by Fusion 360 as a mix of Bonded, Sliding and Separation contact sets (symmetric penetration type), as partially shown below; the complete list is provided in the attached Study Report. The finite element mesh itself (526,544 nodes; 332,937 parabolic solid elements) is shown in Figure 15 (Section 15).

Contact Set	Contact Type	Penetration Type
[M] Sliding7	Sliding	Symmetric
[M] Sliding6	Sliding	Symmetric
[M] Sliding5	Sliding	Symmetric
[M] Sliding4	Sliding	Symmetric
[M] Separation3	Separation	Symmetric
[M] Separation2	Separation	Symmetric
[M] Bonded9	Bonded	Symmetric
[M] Bonded8	Bonded	Symmetric
Bonded54	Bonded	Symmetric
Bonded53	Bonded	Symmetric
Bonded52	Bonded	Symmetric
Bonded51	Bonded	Symmetric
Bonded50	Bonded	Symmetric

Figure 20. Extract of the automatically generated contact sets (Bonded, Sliding and Separation types) from the Fusion 360 Study Report.